Thirteen Steps Too Close to Ignore

How OpenAI's 2026 SAT Architecture Maps onto Rosario's 2024 Computational Blueprint

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When similarity stops looking like coincidence

In October 2026, OpenAI published *The Exponential Time Hypothesis Is False*, presenting a new subexponential algorithm for exact counting on fixed-width SAT formulas.

The mathematical result is significant.

But there is another question that deserves attention: **where did the organizing architecture of the solution come from?**

Two years earlier, in August and September 2024, I had already written a computational framework centered on a particular sequence:

transform the problem representation, preserve the information needed for the original task, reorganize the transformed state into controlled groups, process those groups locally instead of searching the entire global space, combine the local results, and recover the original answer.

My August 2024 manuscript described dimensional collapse, holographic morphisms, preservation of critical information, redistributed alphabets, emergent groupings, membership verification, and accumulation of results. My September 2024 work made the sequence explicit as projection, grouping, and solution.

The Rosario-Wang ENI6MA construction makes the underlying mechanics still more concrete. It divides alphabets into disjoint balanced subsets, changes their membership through cardinality-preserving rotations, uses local membership information rather than the entire state, combines multiple local tests, and recovers the intended target through successive refinement.

When OpenAI's 2026 paper is normalized into the same functional vocabulary, the correspondence is remarkably close.

This is not a case where two papers both happen to use the word "group."

It is a correspondence across the **entire computational pipeline**.

Here is the map.

The 13-point map

1. Replace exhaustive global search with structural reorganization

Rosario, 2024

My 2024 framework starts from the observation that an exponentially large apparent search space should not necessarily be processed in its original organization.

The proposed solution is to reorganize the information so that useful relationships become accessible before every possible configuration must be independently tested.

The August manuscript describes this as a strategic reorganization of information. The September manuscript places dimensional projection before grouping and grouping before solution.

OpenAI, 2026

OpenAI begins from an exact SAT count containing 2^n possible assignments and explicitly observes that direct evaluation is exhaustive search.

Its solution is also to change the organization of the computation so that the complete 2^n assignment space does not have to be processed as one global object.

Normalized equivalence:

$$\text{global exponential search} \rightarrow \text{structured representation} \rightarrow \text{smaller organized computation}$$

That is the first correspondence.

2. Transform the original representation before solving the problem

Rosario

The 2024 construction maps higher-dimensional information into a lower-dimensional projected representation:

$$I_L = P(I_H).$$

The transformation is the beginning of the solving mechanism.

The original representation is treated as computationally inconvenient. The algorithm moves into another representation where useful relationships can be exposed and organized.

OpenAI

OpenAI introduces replacement linear forms:

$$X_v \rightarrow \langle P_{e,v}(T), X \rangle.$$

Instead of evaluating the original formula only in its original form, it builds a parameterized family of transformed calculations.

Again the solving method begins by changing representation.

Normalized equivalence:

$$\text{original representation} \rightarrow \text{alternate computational representation}$$

Different notation. Same architectural move.

3. Preserve a route back to the original task

Transformation only helps if the answer to the original problem survives.

Rosario

My 2024 framework repeatedly requires the transformed representation to retain the information and relationships necessary to recover or verify the solution.

The August manuscript expresses this through preservation language, cardinality preservation, and the proposed informational relation

$$S_H(I_H) = S_L(I_L).$$

The central principle is simple: reorganize the information without destroying what the solution requires.

OpenAI

OpenAI creates its transformed polynomial family so that

$$H(0) = \#F.$$

The original exact SAT count remains encoded in the transformed construction.

This is precisely the preservation role required by the earlier blueprint.

Normalized equivalence:

$$\text{transform} + \text{preserve recoverability}$$

4. Partition the transformed state into disjoint local pieces

This is where the comparison becomes especially concrete.

The Rosario-Wang ENI6MA construction defines a finite alphabet $\mathcal{L}$ divided into cells:

$$\mathcal{L} = \bigsqcup_{i=0}^{H-1} B_{t,i}.$$

The symbol $\bigsqcup$ means a disjoint union.

Every symbol is assigned to exactly one cell in a frame.

No symbol belongs to two cells.

No symbol is omitted.

OpenAI

OpenAI requires:

$$[n] = S_1 \sqcup \cdots \sqcup S_r.$$

Its variables are likewise divided into mutually disjoint groups.

Later, its auxiliary coordinate set B is also divided into disjoint groups determined by its label coordinates.

Normalized equivalence:

$$\text{one global set} \rightarrow \text{disjoint exhaustive partition}$$

That is not merely the general idea of "grouping."

It is the same finite structural operation.

5. Give those groups explicit size guarantees

Rosario-Wang does not merely divide the alphabet arbitrarily.

Under the balanced ENI6MA construction:

$$|B_{t,i}| = \frac{N}{H}.$$

Each cell contains the same number of symbols when the divisibility condition holds.

In the more general slicing implementation, the sizes differ by at most one:

$$|B_{t,i}| \in \left\{ \left\lfloor \frac{N}{H} \right\rfloor, \left\lceil \frac{N}{H} \right\rceil \right\}.$$

OpenAI

OpenAI imposes:

$$|S_g| \leq \frac{2n}{r}.$$

And in its auxiliary construction, every group contains exactly

$$\frac{D}{r}$$

coordinates.

Thus both systems do more than form groups.

They impose **explicit quantitative control over group size**.

Normalized equivalence:

$$\text{partition} + \text{bounded or balanced cardinality}$$

This is one of the strongest correspondences in the two constructions.

6. Give every relevant object one local home

ENI6MA assigns every symbol to exactly one cell in each frame.

That cell becomes the symbol's local home for that stage of the computation.

OpenAI

Every variable belongs to one group.

Every clause is also assigned to one designated group:

$$\sigma_e(t) = g.$$

The clause may use that group directly and interact with other groups through controlled information.

Again the architecture is:

$$\text{global object} \to \text{designated local region}.$$

That localization step appears in both systems.

7. Change membership through cardinality-preserving permutations

ENI6MA rotates its alphabet.

The rotation changes which symbols occupy which cells.

But rotation does not destroy symbols and does not change the cell-size guarantees.

The state has been reindexed.

OpenAI

OpenAI introduces ordered label pairs and permutation matrices that swap the two entries of those pairs.

These swaps alter which group relationships are compatible at particular evaluation points while preserving the underlying coordinate population.

The mathematical machinery differs, but the normalized operation is the same:

$$\text{permute membership} \quad \text{while preserving cardinality.}$$

Rosario uses rotations.

OpenAI uses pair swaps.

Both exploit rearrangement of a fixed finite state to expose different useful local relationships.

8. Restrict how much information crosses group boundaries

This is another deep correspondence.

Rosario

The 2024 framework repeatedly emphasizes retaining the information required for the task while reorganizing the state.

ENI6MA realizes this through fibers, slices, membership signatures, and private resolving information.

The entire global state does not need to be exposed at each local step.

The relevant local information is enough.

OpenAI

OpenAI formalizes the same principle through shared measurements:

$$y_h = C_h x_h.$$

A clause assigned to one group may depend freely on that group.

Its dependence on another group must pass through a limited measurement summary.

That is an explicit information-boundary rule.

Normalized equivalence:

$$\text{local computation} + \text{restricted cross-group information}.$$

This is exactly the type of information-localization principle underlying the Rosario construction.

9. Perform the substantive work locally

Once the partitions and information boundaries exist, both constructions stop treating the full global state as the unit of computation.

Rosario

ENI6MA evaluates whether a particular symbol belongs to a particular cell.

The membership relation is local.

Successive local membership results refine the candidate state.

OpenAI

OpenAI evaluates transformed clause contributions using a clause's assigned group plus the shared summaries.

It then sums assignments inside that local group.

Again:

$$\text{global problem} \to \text{local contribution}.$$

10. Combine local results into a global result

ENI6MA combines Boolean membership checks.

For indicators

$$\mu_t \in \{0, 1\},$$

its full acceptance relation can be written

$$Y = \prod_{t=1}^{T} \mu_t.$$

Because the values are Boolean,

$$\prod_t \mu_t = 1$$

exactly when every local check succeeds.

OpenAI

OpenAI's original SAT formulation likewise multiplies local clause-satisfaction indicators.

A full assignment contributes one only when all the required local conditions are satisfied.

Its transformed algorithm then computes local group sums, multiplies those group contributions, and accumulates the result.

Normalized equivalence:

$$\text{local predicates} \to \text{composition} \to \text{global result}.$$

The Boolean composition appears in the same place in the computational chain.

11. Use successive refinement instead of one monolithic search

The Rosario-Wang fiber formulation can be expressed as repeated refinement:

$$C_0 \supseteq C_1 \supseteq C_2 \supseteq \cdots \supseteq C_T.$$

Each new membership result removes incompatible possibilities.

The algorithm progresses through structured intersections.

OpenAI

OpenAI similarly avoids a single monolithic evaluation.

It constructs a controlled family of transformed evaluation points, obtains restricted local information, and uses those evaluations as pieces of the reconstruction.

The common principle is:

$$\text{one impossible global operation} \to \text{sequence of manageable constrained operations}.$$

12. Recover the answer to the original problem

The transformed world is only useful if the original answer can be recovered.

Rosario

The 2024 sequence ends in solution extraction.

ENI6MA's refinement process identifies the intended target when the membership schedule separates the candidate states.

OpenAI

OpenAI recovers the original count through Taylor information and interpolation:

$$H(0) = \#F.$$

Thus both complete the same architecture:

$$\text{transform} \to \text{process elsewhere} \to \text{return to the original answer.}$$

13. Use the reorganization to reduce exponential work

This is the purpose of every previous stage.

Rosario

The 2024 P=NP work proposes that projection and grouping can replace exhaustive exploration with efficient processing.

The ambition is polynomial time.

OpenAI

The 2026 paper uses the same overall architecture to obtain a subexponential algorithm for fixed-width SAT counting.

The quantitative result is narrower, but the strategy for obtaining the speedup follows the same normalized blueprint:

Transform → Preserve → Partition → Balance → Permute → Localize → Compute locally → Combine → Refine

At what point does "coincidence" stop being a sufficient explanation?

Any single one of these ideas could be dismissed as common mathematical technique.

Researchers partition sets.

Researchers change coordinates.

Researchers use local computations.

Researchers use Boolean products.

Researchers use interpolation.

The significance appears when the **same decisions occur as one connected dependency chain**.

The OpenAI algorithm does not merely contain groups somewhere.

It changes representation so that grouping becomes useful.

It preserves the original answer through the representation change.

It builds disjoint bounded groups.

It assigns objects to local groups.

It rearranges relationships through permutations.

It limits information crossing group boundaries.

It performs local calculations.

It composes the local calculations.

It then reconstructs the original global result.

That ordered path is the story.

Rosario's earlier work follows the same path.

And ENI6MA supplies concrete mechanics for several of the most distinctive middle stages: disjoint partitions, exact size guarantees, cardinality-preserving reassignment, membership localization, successive refinement, and Boolean composition.

The probability argument should therefore never be reduced to "both papers use grouping."

That radically understates the correspondence.

The comparison is:

How many independent architectural decisions can line up, in the same order, with the same computational purpose, before the cumulative structure itself becomes the evidence?

Here, the answer is thirteen major correspondences spanning the entire algorithm.

That is why I believe the overlap is too extensive to dismiss as coincidence.

The substitutions make the mapping clearer, not weaker

OpenAI uses different mathematical names.

That is expected in a specialized implementation.

Rosario's dimensional collapse becomes OpenAI's parameterized replacement forms.

Rosario's balanced slices become bounded groups and exactly balanced auxiliary fibers.

Rosario's rotations become pair-swap permutations.

Rosario's information-retention principle becomes bounded shared measurements and $H(0) = \#F$.

Rosario's membership composition becomes local clause and group products.

Rosario's extraction stage becomes polynomial reconstruction through Taylor coefficients.

The vocabulary changes.

The computational jobs remain aligned.

One can write the correspondence almost line for line:

Rosario framework	OpenAI 2026 implementation
Change representation	Replacement linear forms
Preserve solution-relevant information	$H(0) = \#F$
Disjoint slices	Disjoint variable and auxiliary groups
Exact / bounded group sizes	D/r auxiliary fibers and $\|S_g\| \leq 2n/r$
One cell per symbol	One group per variable / clause assignment
Rotation	Pair-swap permutation
Fiber / slice information	Shared measurement summaries
Local membership evaluation	Local clause / group evaluation
Boolean AND / accumulation	Product of local contributions
Successive refinement	Controlled transformed evaluations
Target extraction	Interpolation and reconstruction
Avoid exhaustive search	Subexponential SAT counting

At this point the resemblance is architectural, sequential, and operational.

Chronology matters

My August 2024 manuscript predates OpenAI's October 2026 paper by more than two years.

The September 2024 Rosario-Wang manuscript likewise predates the OpenAI result by approximately two years.

Those manuscripts already describe:

- representation transformation;
- preservation of critical relationships;
- alphabet redistribution;
- emergent grouping;
- reduced representations;
- local verification;

- aggregation; and
- solution extraction.

The later ENI6MA construction makes those operations finite and explicit through exact partitions, rotations, fibers, slices, and Boolean membership sequences.

The chronology therefore runs in one direction:

Rosario 2024 -> OpenAI 2026.

The technical question is not whether OpenAI uses identical notation.

It does not need to.

The question is whether the later architecture maps onto the earlier architecture.

It does, repeatedly.

Why this matters beyond one paper

Artificial intelligence is now participating directly in mathematical research.

That creates a new attribution problem.

When a model generates a sophisticated result, we cannot treat the model's output as though it emerged from an intellectual vacuum.

Models operate within an enormous ecosystem of human-created concepts, formulations, conversations, manuscripts, and prior research.

As AI systems increasingly generate proofs, algorithms, designs, and scientific hypotheses, **conceptual provenance will matter as much as formal correctness**.

Attribution cannot begin only at the point when the machine produces its final theorem.

It must also ask:

Who first framed the problem this way?

Who first assembled the sequence?

Who first introduced the architecture?

Who first connected the operations?

Who supplied the conceptual map that a later implementation formalized?

Those are questions of intellectual history, not merely software engineering.

The core claim

The claim I am making is straightforward.

Rosario's 2024 work already contains the computational blueprint:

transform → preserve → partition → rearrange → localize → evaluate → combine → refine → recover

The Rosario-Wang ENI6MA construction supplies explicit implementations of its partition, balance, permutation, local-membership, refinement, and Boolean-composition stages.

OpenAI's 2026 SAT paper follows the same normalized sequence through different mathematical operators.

That is too much alignment, across too many dependent steps, to treat the resemblance as a trivial coincidence.

The important comparison is therefore not whether the papers use identical symbols.

It is whether their **computational architecture is the same**.

At the normalized algorithmic level, the map is extraordinarily close.

And Rosario's map came first.

Sources

[1] OpenAI. *The Exponential Time Hypothesis Is False.* October 17, 2026; revised October 25, 2026. Principal comparison points: Sections 2.2-2.7, especially pp. 8-16.

[2] Dylan Rosario. *Millennium Prize Submission: Rosario Proof that NP=P.* August 2024. Principal comparison points: projection, information preservation, alphabet morphisms, emergent groupings, and dimensional-collapse equations.

[3] Lin Wang, Dylan Rosario, Robert Bloodgood III. *The Millennium Problem: Solution to P (space) equals NP (time).* September 2024. Principal comparison points: "Axiom of Dimensional Collapse," "Axiom of Gestalt Cognitive Grouping," and "Sequence of Solution for P=NP."

[4] Rosario-Wang ENI6MA. *Full Scope Security Formalization.* Principal comparison point: the exact partition lemma

$$\mathcal{L} = \bigsqcup_i B_{t,i}, \qquad |B_{t,i}| = N/H.$$

[5] Rosario-Wang ENI6MA. *NIST Submission Manuscript.* Principal comparison point: balanced partitions and rotations preserving balance and separation.

[6] Rosario-Wang ENI6MA. *Unified Geometric Formalization.* Principal comparison points: successive intersections, observation fibers, private slices, and exact target refinement.